Same Test, Six Small Steps

Week 10 Tuesday — the code grows, the question stays the same

COMSC 2043 • Discrete Structures and Critical Thinking

October 20, 2026

The ladder: each file is the last one plus one change

1. `v1` four-cell table, typed by hand
2. `v2` the cells become a function of three inputs
3. `v3` add $P(+ \mid C)$ and $P(C \mid +)$ side by side
4. `v4` loop over prevalence — *predict first*
5. `v5` simulate a million people, compare to the table
6. `v6` the world has a subgroup the stated rates ignore

`python3 compare.py v2 v3` shows **only what changed**.

Critical-thinking overlay: answer the question that was asked

Base-rate neglect is not bad arithmetic

It is answering a *different question* than the one asked.

- "90% accurate" answers: *if you have it, does the test catch you?*
- The patient asked: *I tested positive — do I have it?*
- Three habits: **name the question**, **predict then check**, **say what the check cannot check**.

v1 — the table, every number typed by hand

```
true_pos = 90 # 90% of the 100 who have it
false_neg = 10
false_pos = 495 # 5% of the 9,900 who don't
true_neg = 9405
```

	test +	test −
has it	90	10
does not	495	9405

Thinking move: name the question. Which group is the denominator?

v1 → v2 — assumptions become arguments

```
def screening_table(prevalence, sensitivity,
                    specificity, n=10000):
    have_it = n * prevalence
    dont = n - have_it
    return {
        "true_pos": have_it * sensitivity,
        "false_neg": have_it * (1 - sensitivity),
        "false_pos": dont * (1 - specificity),
        "true_neg": dont * specificity,
    }
```

Thinking move: make assumptions visible. Each argument is a claim someone could challenge.

v2 → v3 — two functions, one table

```
def p_pos_given_c(t): # denominator: 'has it' row
    return t["true_pos"] / (t["true_pos"] + t["false_neg"])

def p_c_given_pos(t): # denominator: 'test +' column
    return t["true_pos"] / (t["true_pos"] + t["false_pos"])
```

$$P(+ \mid C) = 0.9000 \qquad P(C \mid +) = 0.1538$$

Thinking move: charity, then precision. The "90%" claim is true — of the *first* one.

v4 — change one input. Predict first.

```
for prevalence in (0.001, 0.01, 0.05, 0.20, 0.50):
    t = screening_table(prevalence, 0.90, 0.95)
    print(prevalence, p_c_given_pos(t))
```

prevalence	$P(C \mid +)$
0.1%	0.018
1%	0.154
5%	0.486
20%	0.818
50%	0.947

Thinking move: predict, then run. The test never changed. Only who was screened.

v5 — a million simulated people

```
random.seed(10)
has_it = random.random() < prevalence
chance_positive = sensitivity if has_it \
                  else (1 - specificity)
```

calculation 0.1538 simulation 0.1521 difference 0.0017

Thinking move: what does agreement check? The *code* matches the table. Nothing more.

v6 — both agree, and both are wrong

```
in_subgroup = random.random() < 0.10
specificity = 0.80 if in_subgroup else 0.95
```

stated rates: 0.1538 real mix: 0.1231

- More simulation runs shrink noise, not assumption error.
- **Thinking move: what would change my mind?** Evidence about the *population*, not another run.

Where this goes in the Decision Gate

Sources	the three inputs in v2
Rules / Assumptions	"applies to this population" — v6 shows it failing
Work	v1, v2, v3
Check	totals (v2), changed prevalence (v4), simulation (v5)
One-Sentence Summary	no stronger than v6 allows

An AI answer is not evidence until you have rerun the numbers yourself.